

Customer Sales Analysis Report

Objective

The objective of this project is to analyze customer sales transaction data using Python, Pandas, and Matplotlib.

Data Cleaning

The dataset was loaded using Pandas. Missing values were removed using the `dropna()` function to ensure data quality before analysis.

Revenue Calculation

A new column named Revenue was created using the formula:

$\text{Revenue} = \text{Quantity} \times \text{Price}$

This revenue value was used for further analysis.

Revenue by Category

The dataset was grouped by product category and the total revenue generated by each category was calculated.

Key Observation: Clothing generated the highest revenue among all product categories.

Top 10 Customers

Customer transactions were grouped using `customer_id` and total revenue was calculated for each customer. The top 10 customers were identified based on total revenue generated.

Monthly Sales Analysis

The `invoice_date` column was converted into datetime format and monthly sales revenue was calculated.

The monthly sales plot shows the revenue trend across different months.

Summary Statistics

Summary statistics including count, mean, standard deviation, minimum value, and maximum value were generated using the `describe()` function.

Conclusion

The analysis successfully identified the highest revenue-generating categories, top customers, and monthly sales trends. Visualizations helped in understanding sales performance and customer behavior.